



OPI AI Studio 配置记录 - Ubuntu (310P 驱动 + MindIE + Daemon + curl 调用接口)

上一篇在 Windows 下完成了 OPI AI Studio 的基础配置，但遇到了影响使用的报错。因此这次切到 Ubuntu 22.04 + Linux 5.15，重新整理一套可复现的部署流程。

一、目标环境与下载准备

310P NPU 驱动包当前适配的推荐系统为 Ubuntu 22.04，内核版本为 Linux 5.15。

Ubuntu 镜像（`ubuntu-22.04.5-desktop-amd64.iso`）可从以下地址下载：

- <https://mirrors.ustc.edu.cn/ubuntu-releases/22.04.5/ubuntu-22.04.5-desktop-amd64.iso>
- <https://mirrors.huaweicloud.com/ubuntu-releases/22.04.5/ubuntu-22.04.5-desktop-amd64.iso>

二、切换并固定 Linux 5.15 内核

1) 检查当前内核

```
uname -r
```

```
never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ uname -r  
6.8.0-110-generic  
never@never-Thunderobot-Mini-PC:~$
```

2) 安装 5.15.0-126 内核与头文件

```
sudo apt update
```

```
sudo apt install -y linux-image-5.15.0-126-generic linux-modules-extra-5.15.0-126-generic
```

```
sudo apt install -y linux-headers-5.15.0-126 linux-headers-5.15.0-126-generic
```



```
never@never-Thunderobot-Mini-PC: ~  
有 382 个软件包可以升级。请执行 'apt list --upgradable' 来查看它们。  
never@never-Thunderobot-Mini-PC:~$ sudo apt install -y linux-image-5.15.0-126-generic linux-modules-extra-5.15.0-126-generic  
正在读取软件包列表... 完成  
正在分析软件包的依赖关系树... 完成  
正在读取状态信息... 完成  
将会同时安装下列软件：  
    linux-modules-5.15.0-126-generic  
建议安装：  
    fdutils linux-doc | linux-source-5.15.0 linux-tools  
    linux-headers-5.15.0-126-generic  
下列【新】软件包将被安装：  
    linux-image-5.15.0-126-generic linux-modules-5.15.0-126-generic  
    linux-modules-extra-5.15.0-126-generic  
升级了 0 个软件包，新安装了 3 个软件包，要卸载 0 个软件包，有 382 个软件包未被升级。  
需要下载 98.2 MB 的归档。  
解压缩后会消耗 481 MB 的额外空间。  
获取:1 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-updates/main amd64 linux-modules-5.15.0-126-generic amd64 5.15.0-126.136 [22.7 MB]  
获取:2 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-updates/main amd64 linux-image-5.15.0-126-generic amd64 5.15.0-126.136 [11.6 MB]  
27% [2 linux-image-5.15.0-126-generic 2,291 kB/11.6 MB 20%] 3,650 kB/s 20秒^[  
33% [2 linux-image-5.15.0-126-generic 9,414 kB/11.6 MB 81%] 3,650 kB/s 18秒
```

```
never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ sudo apt install -y linux-headers-5.15.0-126  
linux-headers-5.15.0-126-generic  
正在读取软件包列表... 完成  
正在分析软件包的依赖关系树... 完成  
正在读取状态信息... 完成  
下列【新】软件包将被安装：  
  linux-headers-5.15.0-126 linux-headers-5.15.0-126-generic  
升级了 0 个软件包，新安装了 2 个软件包，要卸载 0 个软件包，有 382 个软件包未被升级。  
需要下载 15.2 MB 的归档。  
解压缩后会消耗 103 MB 的额外空间。  
获取:1 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-updates/main amd64 linux-headers-5.15.0-126 all 5.15.0-126.136 [12.3 MB]  
获取:2 http://mirrors.tuna.tsinghua.edu.cn/ubuntu jammy-updates/main amd64 linux-headers-5.15.0-126-generic amd64 5.15.0-126.136 [2,868 kB]  
已下载 15.2 MB，耗时 30秒 (514 kB/s)  
正在选中未选择的软件包 linux-headers-5.15.0-126。  
(正在读取数据库 ... 系统当前共安装有 213215 个文件和目录。)  
准备解压 .../linux-headers-5.15.0-126_5.15.0-126.136_all.deb ...  
正在解压 linux-headers-5.15.0-126 (5.15.0-126.136) ...  
正在选中未选择的软件包 linux-headers-5.15.0-126-generic。  
准备解压 .../linux-headers-5.15.0-126-generic_5.15.0-126.136_amd64.deb ...  
正在解压 linux-headers-5.15.0-126-generic (5.15.0-126.136) ...  
正在设置 linux-headers-5.15.0-126 (5.15.0-126.136) ...
```

3) 修改 GRUB 默认启动项

先安装编辑器：

```
sudo apt install -y vim
```

```
never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ sudo apt install -y vim  
正在读取软件包列表... 完成  
正在分析软件包的依赖关系树... 完成  
正在读取状态信息... 完成  
将会同时安装下列软件：  
    vim-common vim-runtime vim-tiny  
建议安装：  
    ctags vim-doc vim-scripts indent  
下列【新】软件包将被安装：  
    vim vim-runtime  
下列软件包将被升级：  
    vim-common vim-tiny  
升级了 2 个软件包，新安装了 2 个软件包，要卸载 0 个软件包，有 380 个软件包未被  
升级。  
需要下载 9,344 kB 的归档。  
解压缩后会消耗 37.6 MB 的额外空间。  
0% [执行中]
```

编辑 GRUB:

```
sudo vim /etc/default/grub
```

将 `GRUB_DEFAULT` 指向 5.15 内核项:

```
GRUB_DEFAULT="Advanced options for Ubuntu>Ubuntu, with Linux 5.15.0-126-generic"
```

如果推理期间 NPU 利用率和显存占用有明显变化，说明任务已正确落到设备侧。加载进度跑到 100% 后，通常就会继续进入模型推理阶段:

```
never@never-Thunderobot-Mini-PC: ~  
# If you change this file, run 'update-grub' afterwards to update  
# /boot/grub/grub.cfg.  
# For full documentation of the options in this file, see:  
#   info -f grub -n 'Simple configuration'  
  
GRUB_DEFAULT="Advanced options for Ubuntu>Ubuntu,with Linux 5.15.0-126-generic"  
GRUB_TIMEOUT_STYLE=hidden  
GRUB_TIMEOUT=10  
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`  
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash modprobe.blacklist=thunderbolt"  
GRUB_CMDLINE_LINUX=""  
  
# Uncomment to enable BadRAM filtering, modify to suit your needs  
# This works with Linux (no patch required) and with any kernel that obtains  
# the memory map information from GRUB (GNU Mach, kernel of FreeBSD ...)  
#GRUB_BADRAM="0x01234567,0xfefefefe,0x89abcdef,0xefefefef"  
  
# Uncomment to disable graphical terminal (grub-pc only)  
#GRUB_TERMINAL=console  
  
:wq
```

更新并重启:

```
sudo update-grub  
sudo reboot
```

```
never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ sudo update-grub  
Sourcing file `/etc/default/grub'  
Sourcing file `/etc/default/grub.d/init-select.cfg'  
Generating grub configuration file ...  
Found linux image: /boot/vmlinuz-6.8.0-110-generic  
Found initrd image: /boot/initrd.img-6.8.0-110-generic  
Found linux image: /boot/vmlinuz-6.8.0-40-generic  
Found initrd image: /boot/initrd.img-6.8.0-40-generic  
Found linux image: /boot/vmlinuz-5.15.0-126-generic  
Found initrd image: /boot/initrd.img-5.15.0-126-generic  
Memtest86+ needs a 16-bit boot, that is not available on EFI, exiting  
Warning: os-prober will be executed to detect other bootable partitions.  
Its output will be used to detect bootable binaries on them and create new boot entries.  
Found Windows Boot Manager on /dev/nvme0n1p2@/EFI/Microsoft/Boot/bootmgfw.efi  
Adding boot menu entry for UEFI Firmware Settings ...  
done  
never@never-Thunderobot-Mini-PC:~$
```

重启后再次确认：

```
uname -r
```

预期输出： 5.15.0-126-generic

```
never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ uname -r  
5.15.0-126-generic  
never@never-Thunderobot-Mini-PC:~$
```

4) 可选：USB4/雷电 4 机器禁用 thunderbolt 模块

如果主机使用 USB4 或雷电 4 口，连接 OPi AI Studio 启动时可能出现 PCIe 相关错误，可追加以下配置：

```
sudo vim /etc/default/grub
```

```
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash modprobe.blacklist=thunderbolt"
```

```
sudo update-grub
```

```
sudo reboot
```

三、检查硬件识别

系统启动后等待片刻，执行：

```
lspci
```

若能看到 ASMedia 相关信息和 OPi AI Studio PCIe 设备，则链路正常。

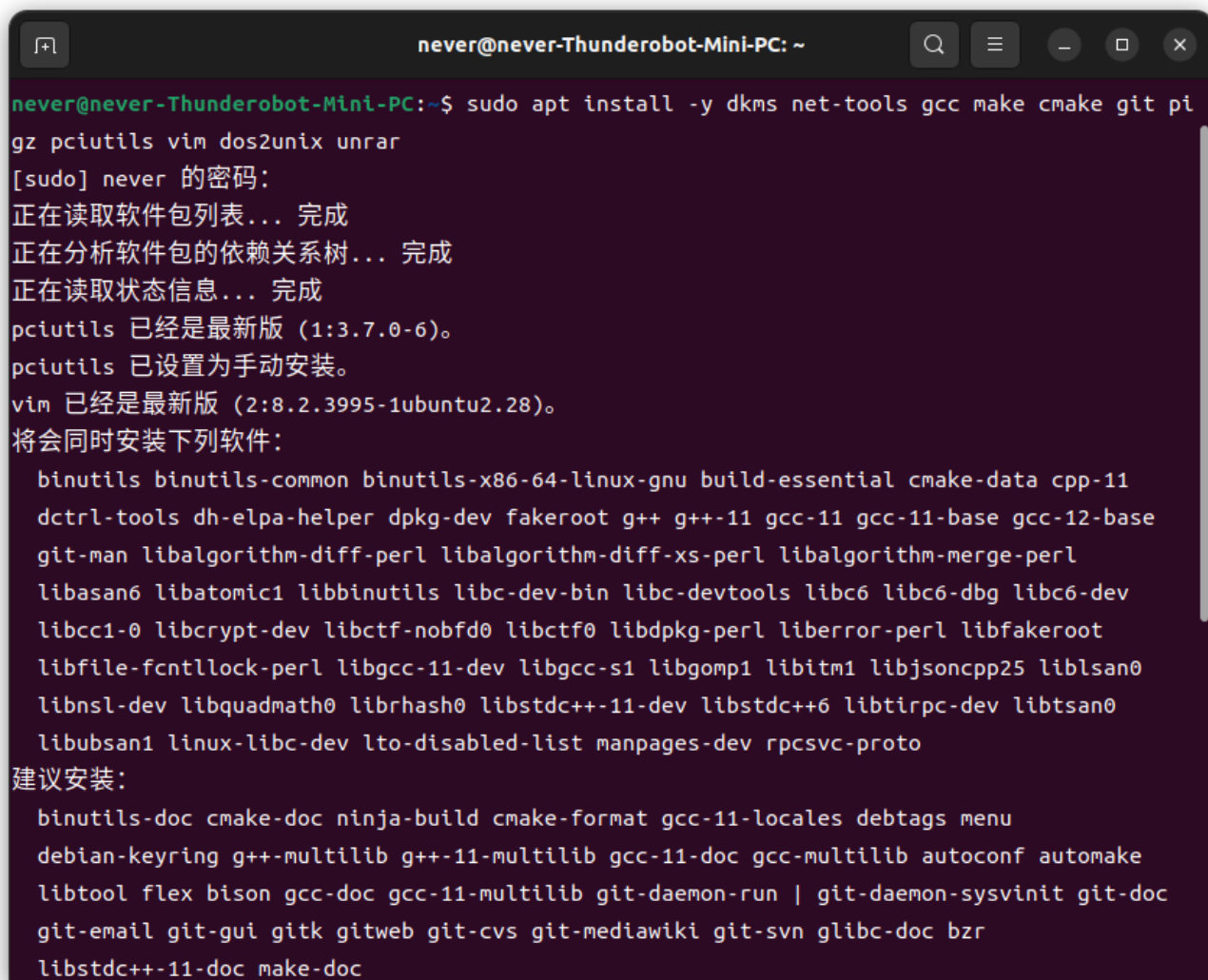


```
never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ lspci | grep ASMedia  
02:00.0 PCI bridge: ASMedia Technology Inc. Device 2464  
03:00.0 PCI bridge: ASMedia Technology Inc. Device 2464  
never@never-Thunderobot-Mini-PC:~$ lspci | grep Huawei  
04:00.0 Processing accelerators: Huawei Technologies Co., Ltd. Device d500 (rev 23)  
never@never-Thunderobot-Mini-PC:~$
```

注意：和 Windows 场景相同，建议先给 OPi AI Studio 上电并连接，再启动电脑；反过来可能导致设备识别异常。

四、安装基础依赖

```
sudo apt update
sudo apt install -y dkms net-tools gcc make cmake git pigz pciutils vim dos2unix unrar
```



```
never@never-Thunderobot-Mini-PC: ~$ sudo apt install -y dkms net-tools gcc make cmake git pi
gz pciutils vim dos2unix unrar
[sudo] never 的密码:
正在读取软件包列表... 完成
正在分析软件包的依赖关系树... 完成
正在读取状态信息... 完成
pciutils 已经是最新版 (1:3.7.0-6)。
pciutils 已设置为手动安装。
vim 已经是最新版 (2:8.2.3995-1ubuntu2.28)。
将会同时安装下列软件:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential cmake-data cpp-11
  dctrl-tools dh-elpa-helper dpkg-dev fakeroot g++ g++-11 gcc-11 gcc-11-base gcc-12-base
  git-man libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl
  libasan6 libatomic1 libbinutils libc-dev-bin libc-devtools libc6 libc6-dbg libc6-dev
  libcc1-0 libcrypt-dev libctf-nobfd0 libctf0 libdpkg-perl liberror-perl libfakeroot
  libfile-fcntllock-perl libgcc-11-dev libgcc-s1 libgomp1 libitm1 libjsoncpp25 liblsan0
  libnsl-dev libquadmath0 librtmp1 libstdc++-11-dev libstdc++6 libtirpc-dev libtsan0
  libubsan1 linux-libc-dev lto-disabled-list manpages-dev rpcsvc-proto
建议安装:
  binutils-doc cmake-doc ninja-build cmake-format gcc-11-locales debtags menu
  debian-keyring g++-multilib g++-11-multilib gcc-11-doc gcc-multilib autoconf automake
  libtool flex bison gcc-doc gcc-11-multilib git-daemon-run | git-daemon-sysvinit git-doc
  git-email git-gui gitk gitweb git-cvs git-mediawiki git-svn glibc-doc bzip2
  libstdc++-11-doc make-doc
```

五、安装 310P NPU 驱动与固件

1) 安装驱动

驱动包: Ascend-hdk-310p-npu-driver_24.1.t29_linux-x86-64-opiaistudo-20250212.run

下载地址: <https://pan.baidu.com/s/1b1hkpebtRXxawnoKQSV--Q?pwd=2m6f>

赋予执行权限：

```
chmod +x Ascend-hdk-310p-npu-driver_24.1.t29_linux-x86-64-opiaistudo-20250212.run
```



A terminal window titled 'never@never-Thunderobot-Mini-PC: ~/桌面/固件和驱动包/npu_driver'. The user enters the command 'chmod +x Ascend-hdk-310p-npu-driver_24.1.t29_linux-x86-64-opiaistudo-20250313.run'. Then, they enter 'll', which displays the following output:

```
总计 717348
drwxr-xr-x 2 never never      4096  4月 28 08:54 ./
drwxr-xr-x 4 never never      4096  4月 28 08:54 ../
-rwxr-xr-x 1 never never 734549807  4月 28 08:53 Ascend-hdk-310p-npu-driver_24.1.t29_
linux-x86-64-opiaistudo-20250313.run*
never@never-Thunderobot-Mini-PC:~/桌面/固件和驱动包/npu_driver$
```

安装（将 `test` 替换为实际普通用户名/组名）：

```
sudo ./Ascend-hdk-310p-npu-driver_24.1.t29_linux-x86-64-opiaistudo-20250212.run \
    --full --install-username=test --install-usergroup=test --install-for-all
```

```
never@never-Thunderobot-Mini-PC: ~/桌面/固件和驱动包/npu_driver
never@never-Thunderobot-Mini-PC:~/桌面/固件和驱动包/npu_driver$ ll
总计 717348
drwxr-xr-x 2 never never      4096  4月 28 08:54 ./
drwxr-xr-x 4 never never      4096  4月 28 08:54 ../
-rwxr-xr-x 1 never never 734549807  4月 28 08:53 Ascend-hdk-310p-npu-driver_24.1.t29_linux-x86-64-opiaistudo-20250313.run*
never@never-Thunderobot-Mini-PC:~/桌面/固件和驱动包/npu_driver$ sudo ./Ascend-hdk-310p-npu-driver_24.1.t29_linux-x86-64-opiaistudo-20250313.run --full --install-username=never --install-usergroup=never --install-for-all
[sudo] never 的密码:
Verifying archive integrity... 100%  SHA256 checksums are OK. All good.
Uncompressing ASCEND DRIVER RUN PACKAGE 100%
[Driver] [2026-04-28 15:36:06] [INFO]Start time: 2026-04-28 15:36:06
[Driver] [2026-04-28 15:36:06] [INFO]LogFile: /var/log/ascend_seclog/ascend_install.log
[Driver] [2026-04-28 15:36:06] [INFO]OperationLogFile: /var/log/ascend_seclog/operation.log
[Driver] [2026-04-28 15:36:06] [INFO]base version is none.
[Driver] [2026-04-28 15:36:06] [WARNING]Do not power off or restart the system during the installation/upgrade
[Driver] [2026-04-28 15:36:06] [INFO]set username and usergroup, never:never
/usr/local/Ascend/driver/tools/upgrade-tool: error while loading shared libraries: libdrvdsmi_host.so: cannot open shared object file: No such file or directory
[Driver] [2026-04-28 15:36:28] [INFO]driver install type: DKMS
[Driver] [2026-04-28 15:36:28] [INFO]upgradePercentage:10%
[Driver] [2026-04-28 15:36:30] [INFO]upgradePercentage:30%
[Driver] [2026-04-28 15:36:31] [INFO]upgradePercentage:40%
[Driver] [2026-04-28 15:36:57] [INFO]upgradePercentage:90%
[Driver] [2026-04-28 15:37:12] [INFO]upgradePercentage:100%
[Driver] [2026-04-28 15:37:12] [INFO]Driver package installed successfully! Reboot needed for installation/upgrade to take effect!
[Driver] [2026-04-28 15:37:12] [INFO]End time: 2026-04-28 15:37:12
never@never-Thunderobot-Mini-PC:~/桌面/固件和驱动包/npu_driver$
```

重启并验证:

```
sudo reboot
npu-smi info
```

```
never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ lspci | grep Huawei  
04:00.0 Processing accelerators: Huawei Technologies Co., Ltd. Device d500 (rev 23)  
never@never-Thunderobot-Mini-PC:~$ npu-smi info  
+-----+  
| npu-smi v1.0 | Version: 24.1.t29 |  
+-----+  
+-----+  
| NPU   Name       | Health   | Power(W)  Temp(C)    Hugepages-Usage(page) |  
| Chip  Device     | Bus-Id   | AICore(%)  Memory-Usage(MB)      |  
+-----+  
| 768   310P1      | OK       | 43.6       47          0 / 0              |  
| 0      0         | 0000:04:00.0 | 0          1924 / 89575      |  
+-----+  
+-----+  
| NPU   Chip       | Process id | Process name | Process memory(MB) |  
+-----+  
| No running processes found in NPU 768 |  
+-----+  
never@never-Thunderobot-Mini-PC:~$
```

2) 安装固件

固件包: Ascend-hdk-310p-npu-firmware_7.6.t7.0.b052-opiaistudio-20250311.run

```
chmod +x Ascend-hdk-310p-npu-firmware_7.6.t7.0.b052-opiaistudio-20250311.run  
sudo ./Ascend-hdk-310p-npu-firmware_7.6.t7.0.b052-opiaistudio-20250311.run --full
```

六、安装 Docker 与 MindIE 镜像

1) 安装 Docker

```
sudo apt install -y docker.io
```

```
never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ sudo apt install docker.io  
[sudo] never 的密码:  
正在读取软件包列表... 完成  
正在分析软件包的依赖关系树... 完成  
正在读取状态信息... 完成  
下列软件包是自动安装的并且现在不需要了:  
  linux-headers-6.8.0-40-generic linux-hwe-6.8-headers-6.8.0-40 linux-hwe-6.8-tools-6.8.0-40  
  linux-image-6.8.0-40-generic linux-modules-6.8.0-40-generic linux-modules-extra-6.8.0-40-generic  
  linux-tools-6.8.0-40-generic  
使用'sudo apt autoremove'来卸载它(它们)。  
将会同时安装下列软件:  
  bridge-utils containerd runc ubuntu-fan  
建议安装:  
  ifupdown aufs-tools btrfs-progs cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2  
  docker-doc rinse zfs-fuse | zfsutils  
下列【新】软件包将被安装:  
  bridge-utils containerd docker.io runc ubuntu-fan  
升级了 0 个软件包，新安装了 5 个软件包，要卸载 0 个软件包，有 293 个软件包未被升级。  
需要下载 79.5 MB 的归档。  
解压缩后会消耗 299 MB 的额外空间。
```

切换 root:

```
sudo -i
```

2) 拉取 MindIE 镜像

```
docker pull --platform=amd64 swr.cn-south-1.myhuaweicloud.com/ascendhub/mindie:3.0.0-300I-
```

OPi AI Studio 配置记录 - L X

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mindie

https://www.hiascend.com/developer/ascendhub/detail/af85b724a7e5469ebd7ea13

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版本	文件大小	更新时间	架构	操作
3.0.0-800I-A3-py311-openeuler24.03-lts	15 GB	2026/05/06	arm64 x86_64	立即下载
3.0.0-800I-A2-py311-openeuler24.03-lts	15 GB	2026/05/06	arm64 x86_64	立即下载
3.0.0-300I-Duo-py311-openeuler24.03-lts	15 GB	2026/05/06	arm64 x86_64	立即下载
3.0.0b2-800I-A2-py311-openeuler24.03-lts	16 GB	2026/04/21	arm64 x86_64	立即下载
3.0.0b2-300I-Duo-py311-openeuler24.03-lts	16 GB	2026/04/21	arm64 x86_64	立即下载
2.3.1-800I-A3-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
2.3.1-800I-A2-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
2.3.1-300I-Duo-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
2.3.0-300I-Duo-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载

mindie

公开

最新版本: 3.0.0-800I-A3-py311-openeuler24.03-lts

推理镜像

大语言模型

镜像概述

镜像下载

mindie镜像版本

版本	大小	更新时间	架构	操作
3.0.0-800I-A3-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
3.0.0-800I-A2-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
3.0.0-300I-Duo-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
3.0.0b2-800I-A2-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
3.0.0b2-300I-Duo-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
2.3.1-800I-A3-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
2.3.1-800I-A2-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
2.3.1-300I-Duo-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载
2.3.0-300I-Duo-py311-openeuler24.03-lts	13 GB	2026/04/16	arm64 x86_64	立即下载

镜像下载

☒ 下载并使用该镜像即表示您同意完全遵守《[镜像许可协议](#)》的条款和条件

前提条件:
准备一台计算机, 要求安装的docker版本必须为1.11.2及以上

操作步骤:
Step 1. 以root登录docker所在的虚拟机
Step 2. 下载镜像

arm架构: docker pull --platform=arm64 swr.cn-south-1.myhuaweicloud.com/ascendhub/mindie:3.0.0-300I-Duo-py311-openeuler24.03-lts

x86架构: docker pull --platform=amd64 swr.cn-south-1.myhuaweicloud.com/ascendhub/mindie:3.0.0-300I-Duo-py311-openeuler24.03-lts

自适应架构: docker pull swr.cn-south-1.myhuaweicloud.com/ascendhub/mindie:3.0.0-300I-Duo-py311-openeuler24.03-lts

```
root@never-Thunderobot-Mini-PC: ~  
root@never-Thunderobot-Mini-PC:~# docker pull --platform=amd64 swr.cn-south-1.myhuaweicloud.com/ascendhub/mindie:3.0.0b2-300I-Duo-py311-openeuler24.03-lts  
3.0.0b2-300I-Duo-py311-openeuler24.03-lts: Pulling from ascendhub/mindie  
b36001b0b1c6: Pull complete  
552c64b26163: Pull complete  
213c522951d5: Pull complete  
afef5e7208a2: Downloading 1.383GB/3.871GB  
adb5317f68bb: Downloading 1.376GB/2.657GB  
a8ab4eff08a6: Download complete  
8b299360212a: Download complete  
977b0c533e1a: Download complete  
6c0e45038bf3: Download complete  
36bf49900f4: Verifying Checksum  
7a3dab41af2e: Download complete  
7d4a6fb937f9: Download complete  
7082163308cd: Download complete  
411a97c52b28: Download complete  
█
```

查看本地镜像列表：

```
docker images
```

```
root@never-Thunderobot-Mini-PC:~# docker images  
REPOSITORY                                TAG                                IMAGE ID            CREATED            SIZE  
swr.cn-south-1.myhuaweicloud.com/ascendhub/mindie 3.0.0b2-300I-Duo-py311-openeuler24.03-lts 813f7f3b291b       2 weeks ago       25.9GB  
root@never-Thunderobot-Mini-PC:~# █
```

3) 编写并执行容器启动脚本

```
vim start_docker.sh
```

新建 `start_docker.sh`：

输入以下内容


```
IMAGE_ID=$1
NAME=${2:-mindie}

if [ -z "$IMAGE_ID" ]; then
    echo "usage: $0 <IMAGE_ID> [CONTAINER_NAME]"
    exit 1
fi

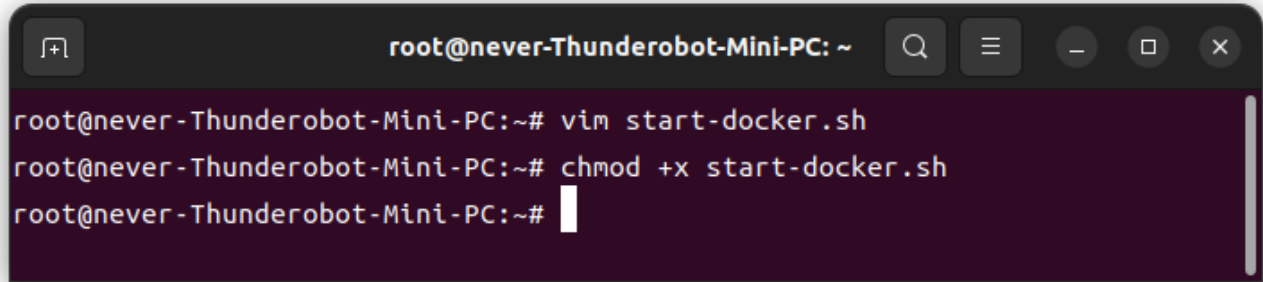
docker run --name "$NAME" -it -d --net=host --shm-size=500g \
    --privileged=true \
    -w /usr/local/Ascend \
    --device=/dev/davinci_manager \
    --device=/dev/hist_hdc \
    --device=/dev/devmm_svm \
    --entrypoint=bash \
    -v /models:/models \
    -v /data:/data \
    -v /usr/local/Ascend/driver:/usr/local/Ascend/driver \
    -v /usr/local/dcmt:/usr/local/dcmt \
    -v /usr/local/bin/npu-smi:/usr/local/bin/npu-smi \
    -v /usr/local/sbin:/usr/local/sbin \
    -v /home:/home \
    -v /tmp:/tmp \
    -v /usr/share/zoneinfo/Asia/Shanghai:/etc/localtime \
    -e http_proxy=$http_proxy \
    -e https_proxy=$https_proxy \
    -e "PATH=/usr/local/python3.11.10/bin:$PATH" \
    "$IMAGE_ID"
```

```
root@never-Thunderobot-Mini-PC: ~  
IMAGE_ID=$1  
NAME=${2:-mindie}  
  
if [ -z "$IMAGE_ID" ]; then  
    echo "usage: $0 <IMAGE_ID> [CONTAINER_NAME]"  
    exit 1  
fi  
  
docker run --name "$NAME" -it -d --net=host --shm-size=500g \  
    --privileged=true \  
    -w /usr/local/Ascend \  
    --device=/dev/davinci_manager \  
    --device=/dev/hist_hdc \  
    --device=/dev/devmm_svm \  
    --entrypoint=bash \  
    -v /models:/models \  
    -v /data:/data \  
    -v /usr/local/Ascend/driver:/usr/local/Ascend/driver \  
    -v /usr/local/dcmt:/usr/local/dcmt \  
    -v /usr/local/bin/npu-smi:/usr/local/bin/npu-smi \  
    -v /usr/local/sbin:/usr/local/sbin \  
    -v /home:/home \  
    -v /tmp:/tmp \  
    -v /usr/share/zoneinfo/Asia/Shanghai:/etc/localtime \  
    -e http_proxy=$http_proxy \  
    -e https_proxy=$https_proxy \  
    -e "PATH=/usr/local/python3.11.10/bin:$PATH" \  
    "$IMAGE_ID"
```

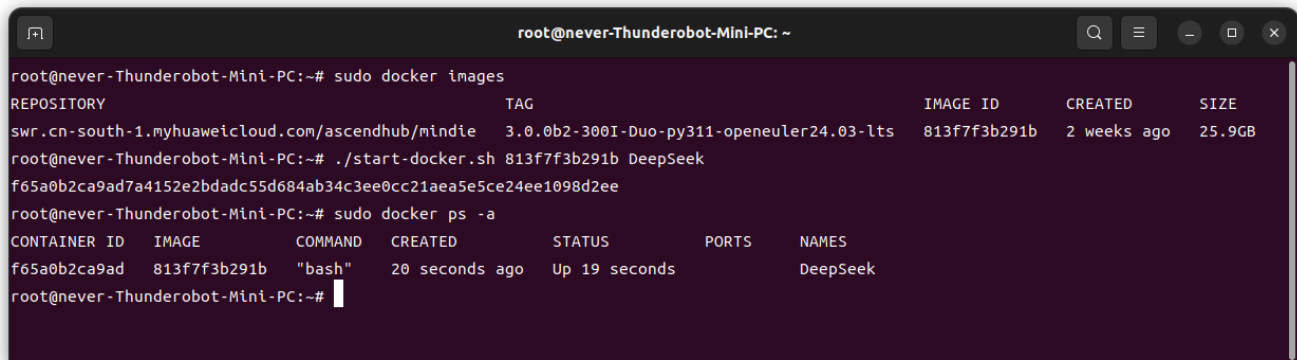
28,13 全部

赋予权限并启动：

```
chmod +x start_docker.sh
./start_docker.sh IMAGE_ID CONTAINER_NAME
```



```
root@never-Thunderobot-Mini-PC: ~
root@never-Thunderobot-Mini-PC:~# vim start-docker.sh
root@never-Thunderobot-Mini-PC:~# chmod +x start-docker.sh
root@never-Thunderobot-Mini-PC:~#
```



```
root@never-Thunderobot-Mini-PC:~# sudo docker images
REPOSITORY                                TAG                                IMAGE ID        CREATED        SIZE
swr.cn-south-1.myhuaweicloud.com/ascendhub/mindie  3.0.0b2-300I-Duo-py311-openeuler24.03-lts  813f7f3b291b   2 weeks ago   25.9GB
root@never-Thunderobot-Mini-PC:~# ./start-docker.sh 813f7f3b291b DeepSeek
f65a0b2ca9ad7a4152e2bdadc55d684ab34c3ee0cc21aea5e5ce24ee1098d2ee
root@never-Thunderobot-Mini-PC:~# sudo docker ps -a
CONTAINER ID   IMAGE                                COMMAND          CREATED          STATUS          PORTS          NAMES
f65a0b2ca9ad   813f7f3b291b                        "bash"          20 seconds ago  Up 19 seconds  -              DeepSeek
root@never-Thunderobot-Mini-PC:~#
```

七、下载模型（DeepSeek-R1-Distill-Qwen-14B）

安装 git-lfs:

```
sudo apt install -y git-lfs
```

```
root@never-Thunderobot-Mini-PC: ~  
root@never-Thunderobot-Mini-PC:~# sudo apt install git-lfs  
正在读取软件包列表... 完成  
正在分析软件包的依赖关系树... 完成  
正在读取状态信息... 完成  
下列软件包是自动安装的并且现在不需要了：  
  linux-headers-6.8.0-40-generic linux-hwe-6.8-headers-6.8.0-40 linux-hwe-6.8-tools-6.8.0-40 linux-  
image-6.8.0-40-generic  
  linux-modules-6.8.0-40-generic linux-modules-extra-6.8.0-40-generic linux-tools-6.8.0-40-generic  
使用'sudo apt autoremove'来卸载它(它们)。  
下列【新】软件包将被安装：  
  git-lfs  
升级了 0 个软件包，新安装了 1 个软件包，要卸载 0 个软件包，有 293 个软件包未被升级。  
需要下载 3,544 kB 的归档。  
解压缩后会消耗 10.5 MB 的额外空间。  
获取:1 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 git-lfs amd64 3.0.2-1ubuntu0  
.3 [3,544 kB]  
7% [1 git-lfs 326 kB/3,544 kB 9%]
```

切换至/models目录：

```
cd /models
```

模型地址：<https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Qwen-14B>

国内网络建议使用镜像：

```
git clone --depth=1 https://modelers.cn/XLRJ/DeepSeek-R1-Distill-Qwen-14B
```

```
root@never-Thunderobot-Mini-PC: /models
root@never-Thunderobot-Mini-PC:/models# ls
root@never-Thunderobot-Mini-PC:/models# git clone --depth=1 https://modelers.cn/XLRJ/DeepSeek-R1-Distill-Qwen-14B
正克隆到 'DeepSeek-R1-Distill-Qwen-14B'...
remote: Enumerating objects: 17, done.
remote: Counting objects: 100% (17/17), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 17 (delta 0), reused 0 (delta 0), pack-reused 0
接收对象中: 100% (17/17), 400.55 KiB | 1.03 MiB/s, 完成.
过滤内容: 100% (5/5), 3.51 GiB | 1.39 MiB/s, 完成.
Encountered 3 file(s) that may not have been copied correctly on Windows:
    model-00003-of-000004.safetensors
    model-00001-of-000004.safetensors
    model-00002-of-000004.safetensors

See: `git lfs help smudge` for more details.
root@never-Thunderobot-Mini-PC:/models#
```

```
root@never-Thunderobot-Mini-PC: /models
root@never-Thunderobot-Mini-PC:/models# git clone --depth=1 https://modelers.cn/XLRJ/DeepSeek-R1-Distill-Qwen-14B
正克隆到 'DeepSeek-R1-Distill-Qwen-14B'...
remote: Enumerating objects: 17, done.
remote: Counting objects: 100% (17/17), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 17 (delta 0), reused 0 (delta 0), pack-reused 0
接收对象中: 100% (17/17), 400.55 KiB | 1.03 MiB/s, 完成.
```

下载过程较长，可在模型目录观察大小变化：

```
du -sh
```

```
root@never-Thunderobot-Mini-PC: /models# du -sh
217M    .
root@never-Thunderobot-Mini-PC: /models# du -sh
1.2G    .
root@never-Thunderobot-Mini-PC: /models#
```

当然也可以通过watch命令持续监控：

```
watch -n 1 du -sh
```

```
root@never-Thunderobot-Mini-PC: /models
Every 1.0s: du -sh          never-Thunderobot-Mini-PC: Thu May  7 18:29:22 2026
4.8G    .
```

八、模型配置、推理验证与服务启动

下面把后续实操整理成一条更顺的链路：先把模型配置和权限处理好，再做一次本地推理验证，最后启动 MindIE 服务并用 `curl` 复核接口。

1) 模型配置调整 (float16)

进入模型目录：

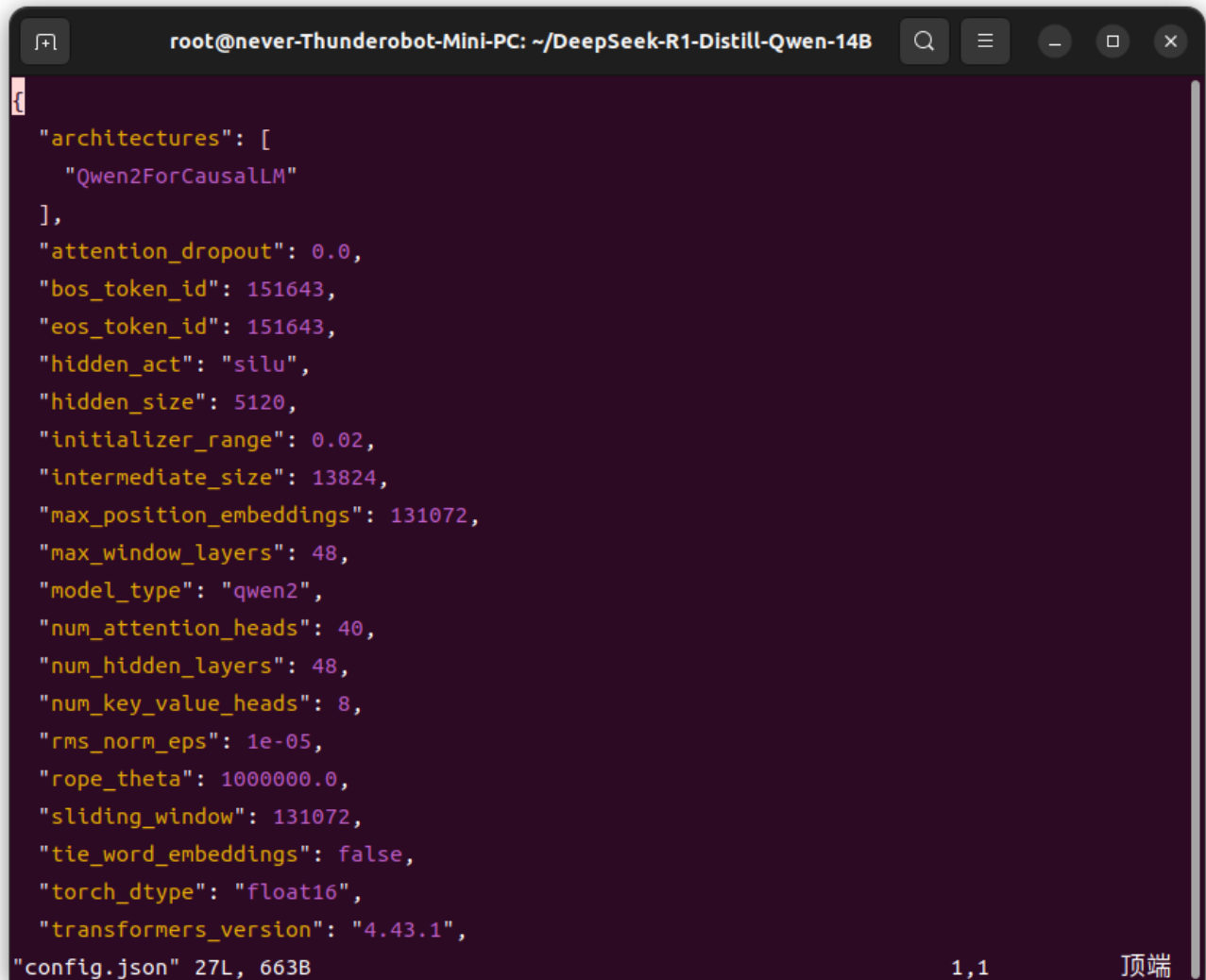
```
cd /models/DeepSeek-R1-Distill-Qwen-14B/
```

编辑配置文件：

```
vim config.json
```

确认或修改 `torch_dtype` 为:

```
"torch_dtype": "float16"
```



The screenshot shows a terminal window with a dark background. The title bar indicates the user is root@never-Thunderobot-Mini-PC and the current directory is ~/DeepSeek-R1-Distill-Qwen-14B. The vim editor is open to config.json, showing a JSON configuration for a Qwen2ForCausalLM model. The configuration includes parameters like attention_dropout, bos_token_id, eos_token_id, hidden_act, hidden_size, initializer_range, intermediate_size, max_position_embeddings, max_window_layers, model_type, num_attention_heads, num_hidden_layers, num_key_value_heads, rms_norm_eps, rope_theta, sliding_window, tie_word_embeddings, torch_dtype (set to float16), and transformers_version. The status line at the bottom of the editor shows "config.json" 27L, 663B, the cursor position 1,1, and the word "顶端" (Top).

```
{
  "architectures": [
    "Qwen2ForCausalLM"
  ],
  "attention_dropout": 0.0,
  "bos_token_id": 151643,
  "eos_token_id": 151643,
  "hidden_act": "silu",
  "hidden_size": 5120,
  "initializer_range": 0.02,
  "intermediate_size": 13824,
  "max_position_embeddings": 131072,
  "max_window_layers": 48,
  "model_type": "qwen2",
  "num_attention_heads": 40,
  "num_hidden_layers": 48,
  "num_key_value_heads": 8,
  "rms_norm_eps": 1e-05,
  "rope_theta": 1000000.0,
  "sliding_window": 131072,
  "tie_word_embeddings": false,
  "torch_dtype": "float16",
  "transformers_version": "4.43.1",
"config.json" 27L, 663B                                     1,1      顶端
```

建议执行一次快速校验，确保 JSON 结构未被破坏：

```
python3 -m json.tool config.json >/dev/null && echo "config.json is valid"
```

```
root@never-Thunderobot-Mini-PC: ~/DeepSeek-R1-Distill-Qwen-14B
root@never-Thunderobot-Mini-PC:~/DeepSeek-R1-Distill-Qwen-14B# python3 -m json.tool config.json >/dev/null && echo "config.json is valid"
config.json is valid
root@never-Thunderobot-Mini-PC:~/DeepSeek-R1-Distill-Qwen-14B#
```

2) 调整模型权限与所有者

在模型目录执行：

```
sudo chmod 640 config.json
cd ..
sudo chown -R root:root /models/DeepSeek-R1-Distill-Qwen-14B/
```

校验权限：

```
ls -l /models/DeepSeek-R1-Distill-Qwen-14B/config.json
ls -ld /models/DeepSeek-R1-Distill-Qwen-14B
```

目标是保证配置文件权限收敛、目录所有者统一，减少容器内运行时的权限歧义。

3) 进入容器并执行一次本地推理

启动并进入容器（`NAME` 为你自定义的容器名）：

```
docker start NAME
docker exec -it NAME bash
```



```
@never-Thunderobot-Mini-PC:/usr/local/Ascend
root@never-Thunderobot-Mini-PC:~# docker start DeepSeek
DeepSeek
root@never-Thunderobot-Mini-PC:~# docker exec -it DeepSeek bash

Welcome to 5.15.0-126-generic

System information as of time: Thu May 7 18:01:20 CST 2026

System load:      1.64
Memory used:      2.6%
Swap used:        0.0%
Usage On:         28%
Users online:     0

[root@never-Thunderobot-Mini-PC Ascend]#
```

进入工作目录：

```
cd atb-models/
```

先执行环境变量配置：

```
export MINDIE_LOG_TO_STDOUT=1
```

再执行推理命令：

```
torchrun --nproc_per_node 1 --master_port 20037 -m examples.run_pa --model_path /models/D
```

另开一个终端观察 NPU 使用情况：

```
watch -n 0.2 npu-smi info
```

Every 0.2s: npu-smi info

npu-smi v1.0		Version: 24.1.t29			
NPU	Name	Health	Power(W)	Temp(C)	Hugepages-Usage(page)
Chip	Device	Bus-Id	AICore(%)	Memory-Usage(MB)	
768	310P1	OK	32.7	40	34402 / 34402
0	0	0000:04:00.0	0	70757 / 89575	
NPU	Chip	Process id	Process name	Process memory(MB)	
768	0	23582	mindie_llm_back	68884	

如果推理期间 NPU 利用率和显存占用有明显变化，说明任务已正确落到设备侧。加载进度跑到 100% 后，通常就会继续进入模型推理阶段：

```
@never-Thunderobot-Mini-PC:/usr/local/Ascend/atb-models

[2026-05-07 19:42:43,193] [5840] [140151754442048] [llmmodels] [INFO] [generate.py-1141] : ---
---total req num: 1, infer start-----
[2026-05-07 19:42:55,722] [5840] [140151754442048] [llmmodels] [INFO] [run_pa.py-434] : -----
-----end inference-----
[2026-05-07 19:42:55,722] [5840] [140151754442048] [llmmodels] [INFO] [run_pa.py-611] : Answer
[0]: How does it differ from traditional machine learning?

</think>

Deep learning is a subset of machine learning that uses neural networks with multiple layers (
deep neural networks) to model complex patterns in data. It differs from traditional machine l
earning in that it can automatically learn hierarchical representations of data without manual
feature engineering, often achieving superior performance on tasks like image and speech reco
gnition.< | end_of_sentence | >
[2026-05-07 19:42:55,722] [5840] [140151754442048] [llmmodels] [INFO] [run_pa.py-612] : Genera
te[0] token num: (0, 72)
[root@never-Thunderobot-Mini-PC atb-models]#
```

Tips: 注意！如果不进行环境变量配置，执行推理命令时将会加载至100%后直接异常退出：

```
@never-Thunderobot-Mini-PC:/usr/local/Ascend/atb-models

[root@never-Thunderobot-Mini-PC atb-models]# torchrun --nproc_per_node 1 --master_port 20037 -m examples.run_pa --model_path /models/DeepSeek-R1-Distill-Qwen-14B/ --max_output_length 256
LogLevelDynamicHandler start
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 100% |  Completed | 293/293 [00:17:00:00, 17.08module/s]
```

4) 启动 mindie-service（服务化）

在容器内进入目录：

```
cd ../mindie/latest/mindie-service/
```

编辑配置：

```
vim conf/config.json
```

将关键字段改为单卡 HTTP 场景：

```
"httpsEnabled": false,
"npuDeviceIds": [[0]],
"worldSize": 1,
"modelWeightPath": "/models/DeepSeek-R1-Distill-Qwen-14B/"
```

```
@never-Thunderobot-Mini-PC:/usr/local/Ascend/mindie/latest/mindie-service

"ipAddress" : "127.0.0.1",
"managementIpAddress" : "127.0.0.2",
"port" : 1025,
"managementPort" : 1026,
"metricsPort" : 1027,
"allowAllZeroIpListening" : false,
"maxLinkNum" : 1000,
"httpsEnabled" : false,
"fullTextEnabled" : false,
"tlsCaPath" : "security/ca/",
"tlsCaFile" : ["ca.pem"],
"tlsCert" : "security/certs/server.pem",
"tlsPk" : "security/keys/server.key.pem",
"tlsCrlPath" : "security/certs/",
"tlsCrlFiles" : ["server_crl.pem"],
"managementTlsCaFile" : ["management_ca.pem"],
"managementTlsCert" : "security/certs/management/server.pem",
"managementTlsPk" : "security/keys/management/server.key.pem",
"managementTlsCrlPath" : "security/management/certs/",
"managementTlsCrlFiles" : ["server_crl.pem"],
-- INSERT --
```

```
@never-Thunderobot-Mini-PC:/usr/local/Ascend/mindie/latest/mindie-service

{
  "npuUsageThreshold" : 0
},

"BackendConfig" : {
  "backendName" : "mindieservice_llm_engine",
  "modelInstanceNumber" : 1,
  "npuDeviceIds" : [[0]],
  "tokenizerProcessNumber" : 8,
  "multiNodesInferEnabled" : false,
  "multiNodesInferPort" : 1120,
  "interNodeTLSEnabled" : true,
  "interNodeTlsCaPath" : "security/grpc/ca/",
  "interNodeTlsCaFiles" : ["ca.pem"],
  "interNodeTlsCert" : "security/grpc/certs/server.pem",
  "interNodeTlsPk" : "security/grpc/keys/server.key.pem",
  "interNodeTlsCrlPath" : "security/grpc/certs/",
  "interNodeTlsCrlFiles" : ["server_crl.pem"],
  "kvPoolConfig" : {"backend": "", "configPath": ""},
-- INSERT --
```

```
@never-Thunderobot-Mini-PC:/usr/local/Ascend/mindie/latest/mindie-service

"truncation" : 0,
"ModelConfig" : [
  {
    "modelInstanceType" : "Standard",
    "modelName" : "DeepSeek",
    "modelWeightPath" : "/models/DeepSeek-R1-Distill-Qwen-14B/",
    "worldSize" : 1,
    "cpuMemSize" : 0,
    "npuMemSize" : -1,
    "backendType" : "atb",
    "trustRemoteCode" : false,
    "async_scheduler_wait_time": 120,
    "kv_trans_timeout": 10,
    "kv_link_timeout": 1080,
    "models" : {
      "layerwiseDisaggregatedMasterDeviceNum" : 2,
      "layerwiseDisaggregatedSlaveDeviceNum" : 8
    }
  }
]
-- INSERT --
```

启动服务：

```
./mindieservice_daemon
```

报错：

```
./mindieservice_daemon: error while loading shared libraries: libtorch_cpu.so: cannot open shared object file: No such file or directory
```

```
[root@never-Thunderobot-Mini-PC mindie-service]# ./bin/mindieservice_daemon
./bin/mindieservice_daemon: error while loading shared libraries: libtorch_cpu.so: cannot
open shared object file: No such file or directory
```

这类报错通常说明运行时环境变量还没补齐。先尝试激活下面这些环境：

```
# 配置CANN环境，默认安装在/usr/local目录下
source /usr/local/Ascend/ascend-toolkit/set_env.sh
# 配置加速库环境
source /usr/local/Ascend/nnal/atb/set_env.sh
# 配置模型仓环境变量
source /usr/local/Ascend/atb-models/set_env.sh
source /usr/local/Ascend/mindie/atb/set_env.sh
```

再次启动后，若看到 `Daemon start success!`，说明服务已经正常拉起：

```
@never-Thunderobot-Mini-PC:/usr/local/Ascend/mindie/latest/mindie-service
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 79%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 80%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 82%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 83%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 84%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 86%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 87%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 88%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 90%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 91%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 92%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 94%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 96%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 97%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 98%|          | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 100%|         | Comp
Loading safetensors checkpoint shards in the lazy mode: Loading safetensors checkpoint shards in the lazy mode: 100%|         | Comp
leted | 293/293 [00:16<00:00, 17.35module/s]
LogLevelDynamicHandler start
Daemon start success!
```

可能出现运存不足时，尝试增加交换分区：

```
sudo fallocate -l 64G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
```

```
root@never-Thunderobot-Mini-PC: ~  
root@never-Thunderobot-Mini-PC:~# fallocate -l 64G /swap_model  
chmod 600 /swap_model  
mkswap /swap_model  
swapon /swap_model  
正在设置交换空间版本 1, 大小 = 64 GiB (68719472640 个字节)  
无标签, UUID=946b5219-d095-4964-bc72-149701c575c2  
root@never-Thunderobot-Mini-PC:~# free -h  
              total        used        free      shared  buff/cache   available  
内存:        62Gi         2.2Gi        8.2Gi         52Mi         51Gi         59Gi  
交换:        63Gi           0B        63Gi  
root@never-Thunderobot-Mini-PC:~#
```

在运行结束时可以关闭交换分区：

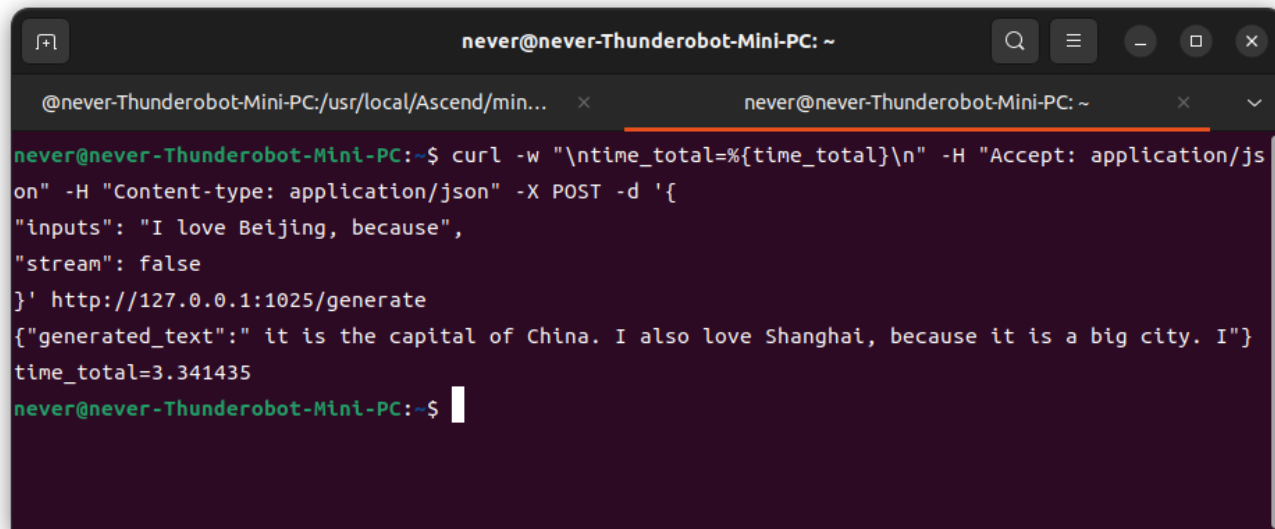
```
sudo swapoff /swapfile  
sudo rm /swapfile
```

5) 使用 curl 验证 HTTP 接口

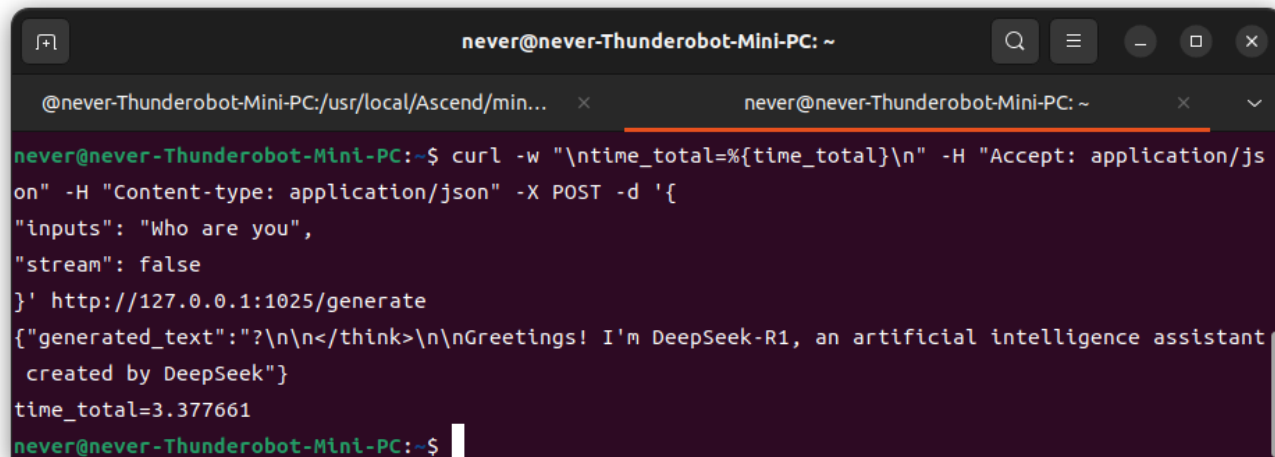
MindIE 服务启动后，可以直接用 `curl` 调用 `generate` 接口做快速验证：

```
curl -w "\ntime_total=%{time_total}\n" \
-H "Accept: application/json" \
-H "Content-type: application/json" \
-X POST -d '{"inputs": "Who are you?", "stream": false}' \
http://127.0.0.1:1025/generate
```

示例结果如下：



```
never@never-Thunderobot-Mini-PC: ~  
@never-Thunderobot-Mini-PC:/usr/local/Ascend/min... x never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ curl -w "\ntime_total=%{time_total}\n" -H "Accept: application/js  
on" -H "Content-type: application/json" -X POST -d '{  
"inputs": "I love Beijing, because",  
"stream": false  
}' http://127.0.0.1:1025/generate  
{"generated_text": " it is the capital of China. I also love Shanghai, because it is a big city. I"}  
time_total=3.341435  
never@never-Thunderobot-Mini-PC:~$
```



```
never@never-Thunderobot-Mini-PC: ~  
@never-Thunderobot-Mini-PC:/usr/local/Ascend/min... x never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ curl -w "\ntime_total=%{time_total}\n" -H "Accept: applicatio/js  
on" -H "Content-type: application/json" -X POST -d '{  
"inputs": "Who are you",  
"stream": false  
}' http://127.0.0.1:1025/generate  
{"generated_text": "?\n\n</think>\n\nGreetings! I'm DeepSeek-R1, an artificial intelligence assistant  
created by DeepSeek"}  
time_total=3.377661  
never@never-Thunderobot-Mini-PC:~$
```



```
never@never-Thunderobot-Mini-PC: ~  
@never-Thunderobot-Mini-PC:/usr/local/Ascend/min... x never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ curl -w "\ntime_total=%{time_total}\n" -H "Accept: application/js  
on" -H "Content-type: application/json" -X POST -d '{  
"inputs": "Tell me something about The Ocean University of China",  
"stream": false  
}' http://127.0.0.1:1025/generate  
{"generated_text": ".\n\n</think>\n\nThe Ocean University of China is a prestigious institution of hi  
gher education located in Qingdao,"}  
time_total=3.348940  
never@never-Thunderobot-Mini-PC:~$
```

```
never@never-Thunderobot-Mini-PC: ~  
@never-Thunderobot-Mini-PC:/usr/local/Ascend/min... x never@never-Thunderobot-Mini-PC: ~  
never@never-Thunderobot-Mini-PC:~$ curl -w "\ntime_total=%{time_total}\n" -H "Accept: application/js  
on" -H "Content-type: application/json" -X POST -d '{  
"inputs": "Who is Elon Musk?",  
"stream": false  
}' http://127.0.0.1:1025/generate  
{"generated_text": "What is he known for?\n\n</think>\n\nElon Musk is a renowned entrepreneur and in  
novator known for"}  
time_total=3.350316  
never@never-Thunderobot-Mini-PC:~$
```

完成了最终接口部署的验证，至此整个 OPi AI Studio 在 Ubuntu 环境下的部署流程就算是完整走通了。

接下来将在此基础之上，进一步调试上下文长度、并发性能等参数，并且换上精度更高的模型进行调试，以实现更快、质量更高的推理能力，尽可能地释放OPi AI Studio的潜力。

同时，在这个基础上通过调用本地部署的模型接口的方式，开始养龙虾(OpenClaw / Hermes Agent)！